ACADEMIC CENTRAL

PRACTICE EXAMINATION

UNIVERSITY · 2ND YEAR

DATABASES

PAPER 1 · PRACTICE A

MARKS: 100 **TIME: 3 hours**

BSc Computer Science, 2nd year, semester 1 final examination. Section A: objective questions (10) · Section B: database concepts, ER modelling, normalisation, SQL, transactions and security (90).

Written by Academic Central for practice, following second-year Computer Science syllabuses at South African universities (database systems). It is not a university examination paper.

INSTRUCTIONS AND INFORMATION

1. This question paper consists of SIX questions. Answer ALL the questions.
2. Write SQL in standard SQL (MySQL or PostgreSQL syntax is accepted).
3. Draw ER diagrams in crow's foot notation.
4. Calculators are NOT allowed.
5. Number your answers exactly as the questions are numbered.

SECTION A

QUESTION 1: MULTIPLE CHOICE

Choose the correct answer. Write only the letter (A–D) next to the question number, e.g. 1.11 B.

1.1 A primary key must be …

A unique and not null.
B a number.
C the first column of the table.
D a foreign key in another table.

1.2 Which SQL statement removes a table and all its data from the database?

A DELETE FROM
B DROP TABLE
C REMOVE TABLE
D TRUNCATE ROW

1.3 Which clause filters groups after GROUP BY?

A WHERE
B HAVING
C ORDER BY
D FILTER

1.4 A foreign key is a column that …

A refers to the primary key of another (or the same) table.
B can never be null.
C is always unique.
D is encrypted.

1.5 Which normal form removes partial dependencies on a composite key?

A 1NF
B 2NF
C 3NF
D BCNF

1.6 In ACID, the letter I stands for …

A Integrity.
B Isolation.
C Indexing.
D Identity.

1.7 Which join returns every row of the left table, even rows with no match in the right table?

A INNER JOIN
B LEFT JOIN
C RIGHT JOIN
D CROSS JOIN

1.8 COUNT(*) counts …

A all rows, including rows with NULL values.
B only rows with no NULL values.
C distinct values only.
D the number of columns.

1.9 One lecturer teaches many modules, and each module has one lecturer. This is a …

A one-to-one relationship.
B one-to-many relationship.
C many-to-many relationship.
D recursive relationship.

1.10 Which of the following is a DDL (data definition language) statement?

A SELECT
B INSERT
C CREATE
D UPDATE

[10]

SECTION B

QUESTION 2: DATABASE CONCEPTS

2.1 What is a database management system (DBMS)? Give TWO advantages of a DBMS over storing data in separate files. (3)

2.2 Explain entity integrity and referential integrity. (4)

2.3 Explain the difference between a candidate key, a primary key and a composite key. (3)

2.4 What is a transaction? Explain what COMMIT and ROLLBACK do. (3)

2.5 What is an index? Give ONE disadvantage of creating many indexes. (2)

[15]

QUESTION 3: ER MODELLING

A community clinic keeps records of patients, doctors and appointments. Each patient has a patient number, name, ID number and phone number. Each doctor has a doctor number, name and speciality. A patient can make many appointments, and each appointment is with exactly one doctor on a date and at a time. A doctor can have many appointments. At an appointment the doctor may prescribe several medicines. A medicine has a code and a name, and can be prescribed at many appointments, each time with its own dosage.

3.1 Identify the FOUR main entities and give a primary key for each. (4)

3.2 Draw an ER diagram in crow's foot notation that shows the entities, their keys and the relationships with their cardinalities. Resolve any many-to-many relationship. (8)

3.3 Write the relational schema for your design (table names with their columns), showing primary keys and foreign keys. (6)

3.4 In which table does the dosage belong? Explain why. (2)

[20]

QUESTION 4: NORMALISATION

A small shop stores its orders in this table:

OrderNo	OrderDate	CustomerNo	CustomerName	ProductCode	ProductName	Qty	UnitPrice
501	2026-03-02	C12	N. Dube	P1	Bread	2	18.00
501	2026-03-02	C12	N. Dube	P4	Milk 2L	1	34.00
502	2026-03-02	C07	S. Pillay	P1	Bread	1	18.00
503	2026-03-03	C12	N. Dube	P9	Eggs (6)	3	27.50

4.1 Using this table, give ONE example each of an insertion anomaly, an update anomaly and a deletion anomaly. (3)

4.2 List the functional dependencies in the table. (4)

4.3 Normalise the table to third normal form (3NF). Give each table with its columns, and underline (or mark PK) the primary keys and mark the foreign keys. (8)

[15]

QUESTION 5: SQL

Answer the questions using the tables below.

Student

studentNo (PK)	name	surname	programme	yearOfStudy
2201	Thabo	Nkosi	CS	2
2202	Lerato	Molefe	CS	2
2203	Pieter	van Wyk	IT	1
2204	Aisha	Patel	CS	3

Module

moduleCode (PK)	moduleName	credits	lecturerId (FK)
COS201	Data Structures	16	L1
COS202	Databases	16	L2
COS203	Operating Systems	16	L1
INF101	Information Systems	12	L3

Enrolment (PK: studentNo, moduleCode, year)

studentNo (FK)	moduleCode (FK)	year	mark
2201	COS201	2026	72
2201	COS202	2026	58
2202	COS201	2026	45
2202	COS203	2026	81

2204	COS201	2026	90
2203	INF101	2026	66

5.1 Write the CREATE TABLE statement for Enrolment, with its primary key, both foreign keys and a CHECK that the mark is from 0 to 100. (5)

5.2 Write an INSERT statement for this: student 2203 has enrolled for COS202 in 2026 and has no mark yet. (2)

5.3 List the name and surname of every CS student, sorted by surname. (3)

5.4 For each module, show the module code and the average mark, but only for modules with an average mark above 60. (4)

5.5 List the student's name, the module name and the mark for every enrolment with a mark of 70 or more. (4)

5.6 Give the result of your query in 5.4 on the data above (before the insert in 5.2). (3)

5.7 Increase every mark in COS202 by 5. (2)

5.8 Delete all enrolments that do not have a mark yet. (2)

5.9 Use a subquery to list the names of students who are NOT enrolled for COS201. (3)

5.10 Explain the difference between WHERE and HAVING. (2)

[30]

QUESTION 6: TRANSACTIONS AND SECURITY

6.1 Explain the lost update problem, using an example of two people updating the same bank balance at the same time. (3)

6.2 Explain how locking prevents the lost update problem. (2)

6.3 What is SQL injection? Give ONE way to prevent it. (3)

6.4 Why must a database be backed up, and how does the transaction log help to recover it after a crash? (2)

[10]

TOTAL: 100

END OF QUESTION PAPER